Haoran Wan

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Education

- **Princeton University**, Ph.D. Candidate in Computer Science, Advisor: Kyle Jamieson — Jul. 2023 - Present
  Teaching Assistant: Distributed Systems '25 Fall, System and Machine Learning '26 Spring
- **Nanjing University**, M.S. in Computer Science, Advisor: Wei Wang — Sep. 2019 - Jun. 2023
  Outstanding Graduate Students Award '21; Principal Special Scholarship '19
- **University of Electronic Science and Technology of China**, B.Eng in Networking Engineering — Sep. 2015 - Jul. 2019
  China National Scholarship '17; 2nd People's Scholarship '16, '18

Skills Summary

- **Languages:** C/C++ (C++14/17, STL, Multithreading), CUDA, Python (PyTorch, NumPy, Pandas), Go, Java, Bash.
- **Systems & Networking:** GPU Systems, Memory/Cache Subsystems, Low-latency Congestion Control, Linux AF_XDP and eBPF.
- **ML Infrastructure:** CUDA Kernel Profiling and Optimization (Hopper, Blackwell), TensorRT, Apache TVM.

Industry Experience

Nvidia | *Software Engineering Internship, Aerial (AI-RAN) Performance Team* — May 2026 - Aug. 2026

- **Aerial**: CUDA Software Performance Optimization | CUDA Kernel and Pipeline Optimization, Memory/Cache Subsystem
  - **Accelerated a production CUDA kernel pipeline by 2.2x** on Hopper and Blackwell GPUs through kernel fusion, warp specialization, and fine-grained parallelism; shipped the changes to the Aerial production codebase.
  - **Evaluated host memory and cache subsystems** under multi-tenant scenarios on the GH200 platform. Mitigated tail latency impact by 63% by leveraging ARM MPAM to partition host CPU cores, L3 cache, and DRAM bandwidth.
  - **Improved the pipeline-level test benches** by including more test cases and implementing a test data tracking infrastructure.

Research Experience

Princeton University | *Advisor: Kyle Jamieson* — July 2023 - Present

- **CellTwin**: ML based Network Algorithm Emulation | Machine Learning, Linux, Congestion Control
  - **Designed and trained a transformer based 5G network emulator** for cross-domain network/application algorithm evaluation, reducing the emulation error of existing methods by more than 55%.
  - **Efficient model deployment** via TensorRT, ONNX and quantization, deployed the ML based emulator with the linux AF_XDP library as a middlebox for plug-and-play real-time packet processing and emulation.
  - **Implemented an automatic data collection** with my NR-Scope tool and reconstructed the commercial network's queue occupancy through an ML based timing model.
- **L4Span**: Ultra-Low Latency Network Stack Optimization | C++, Linux, Congestion Control
  - **Engineered a low-latency zero-copy packet processing module** within an open-source 5G network, utilizing C++ to integrate L4S (Low-Latency, Low-Loss, Scalable) standards and explicit congestion notification (ECN).
  - **Optimized critical data paths** to support advanced congestion control algorithms (TCP Prague, BBRv2&3, SCReAM), reducing packet queuing delay by **95%** while maintaining line-rate throughput.
  - **Designed a real-time capacity predictor** that adjusts queue management strategies based on sub-ms network telemetry.
- **NR-Scope**: High-Throughput Network Telemetry Engine | C++, Multithreading, Real-Time Systems
  - **Architected a real-time telemetry extraction engine** capable of decoding and processing physical layer control signals within a strict **0.5 ms** transmission window, established as a foundational open-source telemetry tool for network researchers.
  - **Engineered a custom C++ thread pool and worker queue system** to parallelize signal decoding, enabling real-time introspection of downlink/uplink resource allocation without blocking the critical path.

Nanjing University | *Advisor: Wei Wang* — Sept. 2019 - Jun. 2023

- **ALT:** End-to-End Deep Learning Compiler | C++, Python, Apache TVM, CUDA
  - **Architected a high-performance ML compiler** that unifies graph-level data layout and operator-level loop optimizations, resolving the "phase ordering" problem in traditional compilation stacks.
  - **Engineered a custom Auto-Tuning Engine** on top of Apache TVM, implementing a cost model that searches for optimal tensor layouts (NCHW/NHWC) and loop tiling strategies simultaneously.
  - **Achieved 1.5x inference speedup** on single operators (Conv2D, MatMul) and 1.4x end-to-end latency reduction on ResNet, BERT, MobileNets models by generating hardware-aware CUDA/LLVM code.

Selected Publications

- **[USENIX NSDI '27]** CellTwin: Reproducible Cellular Network Emulation via High Resolution, in situ Measurement-Driven ML. **Haoran Wan**, Yaxiong Xie, Kyle Jamieson
- **[ACM SIGCOMM '26]** Synchronizing with the Scheduler: Dual-Loop Congestion Control for 5G Uplink on Commodity Devices. Qiang Wu, Yuxin Liu, Tianyang Zhang, **Haoran Wan**, Kyle Jamieson, Yaxiong Xie
- **[ACM CoNEXT '25]** L4Span: Spanning Congestion Signaling over NextG Networks for Interactive Applications. **Haoran Wan**, Kyle Jamieson
- **[ACM IMC '25]** Automated, Cross-Layer Root Cause Analysis of 5G Video-Conferencing Quality Degradation. Fan Yi, **Haoran Wan**, Kyle Jamieson, Oliver Michel
- **[ACM CoNEXT '24]** NR-Scope: A Practical 5G Standalone Telemetry Tool. **Haoran Wan**, Xuyang Cao, Alexander Marder, Kyle Jamieson
- **[ACM EuroSys '23]** ALT: Boosting Deep Learning Performance by Breaking the Wall between Graph and Operator Level Optimizations. Zhiying Xu, Jiafan Xu, Hongding Peng, Wei Wang, Xiaoliang Wang, **Haoran Wan**, Haipeng Dai, Yixu Xu, Hao Cheng, Kun Wang, Guihai Chen